

SAT-Style Practice Test — Original Website Edition

Reading and Writing + Math — All Four Modules

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SECTION 1, MODULE 1: Reading and Writing

Question 1 Skill: Words in Context

The Voyager space probes (1977–present) carried scientific instruments and golden records into deep space, generating vast quantities of data. Astrophysicist Dr. Lena Morales, who studies interstellar plasma, continues to analyze Voyager’s transmissions, demonstrating that the missions’ contributions to science are _____.

Which choice completes the text with the most logical and precise word or phrase?

- A) interrupted
- B) ongoing
- C) unprecedented
- D) debatable

Answer: B

Question 2 Skill: Words in Context (Phrase)

Because zooplankton and other microscopic marine organisms inhabiting coastal waters are consumed by herring as well as by various seabirds and larger fish, zooplankton are _____ coastal marine ecosystems. Declining zooplankton numbers, which may be driven by rising ocean temperatures, thus has a direct, negative effect on oceanic food chains.

Which choice completes the text with the most logical and precise word or phrase?

- A) tangential to
- B) symbolic of
- C) integral to
- D) interchangeable with

Answer: C

Question 3 Skill: Words in Context

Certain novels might carry multiple layers of meaning — the works of Toni Morrison can _____ as many distinct readings as there are people to engage with them — and as long as a reader’s interpretation is not deliberately absurd or born of carelessness, it is hard to argue that the reader has fundamentally misread the text.

Which choice completes the text with the most logical and precise word or phrase?

- A) dismiss
- B) sustain
- C) constrain
- D) obscure

Answer: B

Question 4 Skill: Words in Context

The number of vowel sounds across the world's tonal languages varies considerably, tending toward the ____: Yoruba, spoken across West Africa, employs only seven oral vowels, and Vietnamese, the national language of Vietnam, uses a similarly small set, while the global average number of vowels per language exceeds eleven.

Which choice completes the text with the most logical and precise word or phrase?

- A) minimal
- B) uniform
- C) elaborate
- D) rhythmic

Answer: A

Question 5 Skill: Main Purpose

Established in Washington, D.C., in 1989, the National Museum of African Art is dedicated to the collection, study, and exhibition of traditional and contemporary African art. Since its founding, it has gathered over 12,000 objects for its permanent holdings. A more recently established institution focused on African diaspora art is the Museum of the African Diaspora in San Francisco, California. Located in the Yerba Buena arts district, it emphasizes the cultural connections among peoples of African descent worldwide.

Which choice best states the main purpose of the text?

- A) To explain a historical development that led to the creation of two cultural institutions
- B) To describe how two institutions obtained resources to build their permanent collections
- C) To present information about two institutions, including each institution's area of focus
- D) To compare the collection sizes of two institutions dedicated to a shared subject

Answer: C

Question 6 Skill: Main Purpose (Poetry)

The following text is adapted from William Cullen Bryant's 1821 poem "Green River." The poem reflects on a river the speaker frequently visits in the countryside of western Massachusetts.

Though foul with stain of man's misuse, And choked with ruins of his mill, Thy springs are fresh, thy current views Move outward with a tranquil will, And gently past the grassy lea Thy shining waters wind their way.

Which choice best states the main purpose of the text?

- A) It describes the speaker's frustration with the daily pace of rural life near the river.
- B) It contrasts the natural purity of the river's springs with the pollution caused by human industry.
- C) It emphasizes the danger posed by an incoming flood expected to reach the river's banks.
- D) It compares two different types of watercraft used on the river.

Answer: B

Question 7 Skill: Inference

In what is now Arizona, the Hopi people operate the Hopi Cultural Center on their ancestral lands. Drawing on centuries of oral tradition and community knowledge, the center presents Hopi history and culture to both tribal members and visitors. The Navajo Nation, a neighboring sovereign nation, maintains a similar institution in its own cultural center in Window Rock. Both centers differ from typical state-run museums; when exhibiting Indigenous artifacts, such museums tend to assume primarily non-Indigenous visitors and employ Western curatorial methods for organizing their displays.

According to the text, what is one way that typical state-run museums differ from the cultural centers operated by the Hopi and the Navajo Nation?

- A) The museums typically display a smaller number of artifacts in their galleries.
- B) The museums are generally more compact in their physical footprint.
- C) The museums are primarily designed for non-Indigenous audiences.
- D) The museums focus exclusively on historical rather than contemporary cultural items.

Answer: C

Question 8 Skill: Detail / Comprehension

The Galápagos archipelago is home to the marine iguana (*Amblyrhynchus cristatus*), one of several reptile species found nowhere else on Earth. Due to the Galápagos Islands' uniquely isolated geography and volcanic terrain, the marine iguana evolved highly specialized traits as it adapted to forage in the surrounding ocean. However, like many highly specialized species, the marine iguana is especially susceptible to environmental disturbances, such as oil spills, invasive predators, and El Niño events, that destabilize the fragile island ecosystems in which the animals live.

Based on the text, why is the marine iguana a highly specialized species?

- A) It inhabits only the most geographically remote islands in the Galápagos chain.
- B) It evolved over time to exploit a specific ecological niche found on the Galápagos Islands.
- C) It developed the ability to share its marine habitat with other highly specialized reptile species on the archipelago.
- D) It thrives in an environment exposed to frequent environmental disruptions.

Answer: B

Question 9 Skill: Main Idea (Literature)

The following text is adapted from Virginia Woolf's 1927 essay collection. The narrator describes the experience of walking through a London neighborhood at dusk.

I stepped out of the doorway into the evening air and walked along the embankment. The lamps were being lit one by one, dropping little circles of gold upon the grey pavement. Across the river, the silhouette of the cathedral darkened against a sky still faintly luminous. There was a sense, walking here, that the city had paused — that between the bustle of the day and the theatre of the night, London held its breath, and one could hear, beneath the rumble of traffic, the quiet pulse of the river sliding past the stone.

Which choice best states the main idea of the text?

- A) The narrator contrasts feelings of unease with the apparent calm of the city as evening falls along the river.
- B) The narrator is eager for the evening's entertainment to begin and is impatient with the slowness of twilight.
- C) Although the narrator did not plan to walk through the neighborhood, they are captivated by an unexpected sense of discovery.
- D) The narrator perceives a parallel between the transitional moment of dusk in the city and the quiet rhythm of the river.

Answer: D

Question 10 Skill: Data Interpretation (Graph)

To investigate agricultural consolidation trends in central Brazil, Mariana Costa and colleagues used satellite imagery and census data to track changes in the average farm size across three agricultural land-use categories (categories that indicate the primary use of farmland). Due to the central plateau's favorable climate, large tracts of savanna were converted to farmland as commodity prices rose during the 1970s. By 1995, average farm sizes had dropped considerably. In that year, they were between ____.

[Graph: "Average Farm Size Across Three Land-Use Categories in Central Brazil." Three lines for Category I (large-scale soy, ~125 ha in 1960 dropping to ~30 ha by 1995), Category II (mixed subsistence, ~75 ha

dropping to ~20 ha by 1995), and Category III (cattle ranching, ~100 ha dropping to ~25 ha by 1995). Y-axis 0–150 hectares, X-axis 1960–2010.]

Which choice most effectively uses data from the graph to complete the assertion?

- A) 100 and 125 hectares.
- B) 75 and 100 hectares.
- C) 60 and 80 hectares.
- D) 20 and 30 hectares.

Answer: D

Question 11 Skill: Data Interpretation (Graph)

[Graph: “Butterfly Species Observed in a Southern England Meadow, March 2019 to December 2019.” Three lines: Painted Lady (solid/triangle, starts ~60 in March, drops to near 0 by September/December), Common Blue (dashed/square, steady ~10 all year), Red Admiral (dotted/circle, starts ~2, rises to ~15 by December).]

Keiko Tanaka and colleagues recorded butterfly populations in a temperate meadow. They observed that some species were entirely absent from the meadow at certain times of the year; for example, they did not observe even one _____.

Which choice most effectively uses data from the graph to complete the example?

- A) Common Blue in March of 2019.
- B) Red Admiral in March and June of 2019.
- C) Painted Lady in September and December of 2019.
- D) Common Blue in June of 2019.

Answer: C

Question 12 Skill: Weaken Hypothesis

Geologist Haruto Nakamura and colleagues examined sediment cores from two volcanic lakes, one near Iceland and one near the Azores. Core S-12 from Iceland contained mineral deposits formed during periods of intense volcanic activity, while core S-07 from the Azores showed a chemical signature indicating the sediment settled in water averaging 8°C. Azorean volcanic lakes typically maintain temperatures of at least 12°C. Because water temperature decreases with depth, a geology student hypothesized that the Azorean lake deposits formed in colder water than the Icelandic ones.

Which finding, if true, would most directly weaken the student’s hypothesis?

- A) Icelandic volcanic lakes tend to be replenished by glacial meltwater shortly after eruptions, whereas Azorean lakes are not.
- B) The water temperature at a given depth tends to be higher in Azorean waters than at that depth in Icelandic waters.
- C) There are not enough suitable deep volcanic lakes in Iceland for sediment to be sampled reliably at great depth.
- D) The water temperature at a given depth tends to be lower in Azorean waters than at that depth in Icelandic waters.

Answer: B

Question 13 Skill: Logical Completion

Several studies have documented a positive correlation between soil nitrogen levels and fungal diversity in temperate forests across North America. But Dr. Anika Patel and Dr. Ravi Menon did not observe this correlation in a study conducted in boreal forests of Scandinavia, leading some ecologists to propose that the pattern is unique to North American ecosystems. However, Dr. Yuki Sato and colleagues reported nitrogen and fungal diversity rising or falling in tandem in forests in Japan, and multiple other studies from outside North America have produced similar findings, suggesting that _____.

Which choice most logically completes the text?

- A) environmental conditions unique to Scandinavia likely cause both nitrogen levels and fungal diversity to be significantly higher there than elsewhere.
- B) the forests in Patel and Menon's study may be more similar to the North American forests than to those in Sato's study.
- C) there may be circumstances specific to the boreal forests in Patel and Menon's study that disrupted a general positive association between nitrogen and fungal diversity.
- D) temperate forests outside North America are more likely to show a negative correlation between nitrogen and fungal diversity than a positive one.

Answer: C

Question 14 Skill: Logical Completion

Dr. Elena Vasquez et al. analyzed historical population scaling trends across pre-Columbian settlements in Mesoamerica, focusing on urban growth from approximately 200 CE to 900 CE. They examined settlements across the region, with groupings based on regionally distinct trade networks, religious traditions, and geographic constraints — including the powerful city-state of Teotihuacan in central Mexico, maritime trading centers along the Gulf Coast, and highland market towns in Guatemala. A consistent trend emerged: as a settlement's total inhabited area expanded, so did population density in its central districts. This finding implies that _____.

Which choice most logically completes the text?

- A) even when they were substantial, known inter-regional differences in trade and religious traditions had a limited effect on urban growth patterns in pre-Columbian Mesoamerican settlements.
- B) the observed relationship is heavily influenced by geographic features specific to Mesoamerica, making it unlikely that a similar relationship would be found in other ancient civilizations.
- C) the effects of variations in trade networks on urbanization were more evident in comparisons within a region than in comparisons between regions.
- D) despite regional differences in their trade networks and spiritual practices, pre-Columbian Mesoamerican settlements shared common spatial constraints that encouraged high population density in central urban areas.

Answer: D

Question 15 Skill: Pronoun Agreement

After lengthy deliberation, _____ finally reached a verdict: the committee for the 1953 Pulitzer Prize in Fiction would award that year's honor to Ernest Hemingway for *The Old Man and the Sea*, praising its power, style, and emotional honesty.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) someone
- B) you
- C) they
- D) it

Answer: C

Question 16 Skill: Pronoun Reference

In March of 1958, the album *Johnny's Greatest Hits* by Johnny Mathis became a fixture on the Billboard charts. Having remained on the chart for an extraordinary 490 consecutive weeks, _____ ranked among the longest-charting albums in the history of recorded music.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) those
- B) they

- C) each
- D) it

Answer: D

Question 17 Skill: Sentence Structure

In 1818, the novel *Frankenstein* was published without any author named on the title page. Today, we know that _____.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) was the author's real name Mary Shelley?
- B) the author's real name was Mary Shelley.
- C) was the author's real name Mary Shelley.
- D) the author's real name was Mary Shelley?

Answer: B

Question 18 Skill: Punctuation (Comma after geographic name)

While the Andean condor (*Vultur gryphus*) can be spotted soaring over mountainous terrain in Peru and parts of southern _____ than 70 percent of the remaining population of this endangered raptor inhabits a narrow corridor of the Chilean Andes. There, ornithologist Dr. Catalina Reyes is leading conservation efforts.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Argentina, more
- B) Argentina; more
- C) Argentina. More
- D) Argentina more

Answer: A

Question 19 Skill: Possessives / Plural

India and Brazil have two of the most detailed constitutions in the world. At 146,385 and 64,488 words, respectively, these _____ according to comparative law scholars Mila Versteeg and Emily Zackin, correlate with each nation's approach to embedding social and economic rights directly into foundational legal texts.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) constitutions' lengths,
- B) constitutions lengths,
- C) constitution's length,
- D) constitutions' lengths,

Answer: B

Question 20 Skill: Punctuation (Appositive)

A seated woman gazes pensively at a vase of sunflowers in *Interior with Flowers*, an oil painting by Danish _____ Vilhelm Hammershøi.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) artist:
- B) artist
- C) artist—
- D) artist,

Answer: B

Question 21 Skill: Transitions

Every Canadian province has an official provincial mineral, which is typically nominated by a panel of geologists and then approved through a legislative process. For example, Ontario's amethyst was officially designated in 1975, and British Columbia's jade in 2007. _____ years elapse between a mineral's nomination and official designation, as the parliamentary process can be remarkably slow.

Which choice completes the text with the most logical transition?

- A) Often,
- B) Therefore,
- C) Similarly,
- D) Indeed,

Answer: A

Question 22 Skill: Transitions

A 2005 study of migrating wildebeest observed the herd abruptly changing direction after a few lead animals detected a predator. This is an example of a collective behavior pattern that benefits the group. _____ the animal kingdom is occasionally prone to cascading misjudgments, often with harmful outcomes: a 2012 study of emperor penguins found that following a disoriented lead penguin caused the colony to march away from their breeding grounds unnecessarily.

Which choice completes the text with the most logical transition?

- A) In other words,
- B) On the other hand,
- C) Indeed,
- D) As a result,

Answer: B

Question 23 Skill: Rhetorical Synthesis (Notes → Process Description)

While researching a topic, a student has taken the following notes:

- *Guernica* is a 1937 large-scale oil painting by Spanish artist Pablo Picasso.
- It depicts the horrors of the bombing of the Basque town of Guernica.
- Mural painting is a technique in which an image is applied directly to a wall, ceiling, or other large permanent surface.
- The surface is often prepared with plaster or another base layer.
- *Guernica*, despite its enormous size, was painted on canvas rather than directly on a wall.

The student wants to describe the medium of *Guernica*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Mural painting involves applying an image to a prepared surface such as plaster, not to a canvas.
- B) Picasso used oil paint on canvas to create *Guernica*, a large-scale work depicting the bombing of a Basque town.
- C) Picasso created *Guernica* in response to events during the Spanish Civil War.
- D) In mural painting, an image is applied to a large surface that is often prepared with plaster.

Answer: B

Question 24 Skill: Rhetorical Synthesis (Emphasize Similarity)

While researching a topic, a student has taken the following notes:

- Typically, a country's most visited city is also its capital.

- The most visited city in Turkey is Istanbul.
- Turkey's capital is Ankara.
- The most visited city in Australia is Sydney.
- Australia's capital is Canberra.

The student wants to emphasize a similarity between the two countries. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) In both Turkey and Australia, the most visited city is not the country's capital.
- B) While the most visited city in Turkey is Istanbul, the most visited city in Australia is Sydney.
- C) The most visited city in Turkey is Istanbul, but the country's capital is Ankara.
- D) Turkey's capital is Ankara; Australia's capital is Canberra.

Answer: A

Question 25 Skill: Rhetorical Synthesis (Emphasize Method)

While researching a topic, a student has taken the following notes:

- Journalist Sonia Nazario's book *Enrique's Journey* chronicles immigration from Central America to the United States.
- The Great Migration of Central Americans surged during the late twentieth and early twenty-first centuries.
- To document this phenomenon, Nazario narrates the personal experience of a Honduran teenager named Enrique, who travels through Mexico to find his mother.
- The book won the Grand Prize of the Robert F. Kennedy Book Award in 2007.
- It also received the PEN Center USA Literary Award for nonfiction in 2007.

Which choice most effectively uses information from the notes to emphasize how Nazario documents the immigration phenomenon?

- A) The United States has received large numbers of immigrants from Central America, a subject explored in award-winning journalism.
- B) Sonia Nazario brings the sweeping story of Central American immigration into vivid focus by following one teenager's harrowing journey through Mexico to reunite with his mother.
- C) In 2007, *Enrique's Journey* received both the Robert F. Kennedy Book Award and the PEN Center USA Literary Award.
- D) The journey of Enrique, a Honduran teenager, is the subject of the award-winning book *Enrique's Journey*.

Answer: B

Question 26 Skill: Rhetorical Synthesis (Emphasize Similarity Between People)

While researching a topic, a student has taken the following notes:

- Yo-Yo Ma is a French-born American cellist and composer.
- Critics have praised his playing for its combination of technical precision and deep emotional resonance.
- His acclaimed recordings span many genres and feature numerous guest artists.
- Itzhak Perlman performed violin on Ma's 1998 album, *Inspired by Bach*.
- Emanuel Ax performed piano on Ma's 2001 album, *Soul of the Tango*.

The student wants to emphasize a similarity between Itzhak Perlman and Emanuel Ax. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Itzhak Perlman played on *Inspired by Bach*, which was released three years before *Soul of the Tango*.
- B) *Inspired by Bach* and *Soul of the Tango*, released in 1998 and 2001 respectively, are albums by cellist Yo-Yo Ma.
- C) Although both musicians have performed on Yo-Yo Ma albums, Perlman played violin whereas Ax played piano.

D) Itzhak Perlman and Emanuel Ax both contributed their musical talents to albums by Yo-Yo Ma in the late 1990s and early 2000s.

Answer: D

Question 27 Skill: Rhetorical Synthesis (Generalization About Early Career)

While researching a topic, a student has taken the following notes:

- Sidney Poitier (1927–2022) was an actor born in Miami, Florida, and raised in the Bahamas.
- He was of Bahamian heritage.
- He was the first Black man to win the Academy Award for Best Actor.
- Early in his career, he appeared in many minor, uncredited roles, such as his part in *No Way Out* (1950).
- Later, he starred in many major, critically acclaimed films, such as *In the Heat of the Night* (1967).

The student wants to make and support a generalization about Poitier's early career. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) *No Way Out* (1950) was released early in Poitier's career, whereas *In the Heat of the Night* (1967) came later.
- B) Poitier was not always given prominent roles early on — for instance, in *No Way Out* (1950), he appeared in a minor, uncredited part.
- C) *No Way Out* and *In the Heat of the Night* are both films featuring Poitier, the first Black man to win the Academy Award for Best Actor.
- D) In *No Way Out* (1950), actor Sidney Poitier appeared in a minor, uncredited role.

Answer: B

Question 1 Skill: Words in Context

Despite its stated ambition to represent global perspectives, much prominent research on childhood development published in the 2000s drew disproportionately from studies conducted in North America and Western Europe, partly due to limited institutional funding elsewhere. Researchers would later _____ this shortcoming after securing new partnerships with universities in Southeast Asia, such as Vietnam, and sub-Saharan Africa, such as Kenya.

Which choice completes the text with the most logical and precise word or phrase?

- A) endorse
- B) mitigate
- C) classify
- D) presuppose

Answer: B

Question 2 Skill: Words in Context

Though the bioluminescence mechanisms of deep-sea organisms, such as the anglerfish *Melanocetus johnsonii*, have received limited formal investigation, anecdotal reports have suggested that light production serves primarily as a lure for prey. Experimental results by Dr. Tomoko Hayashi et al. now _____ this idea: captive anglerfish in darkened tanks activated their bioluminescent organs almost exclusively in the presence of potential prey species.

Which choice completes the text with the most logical and precise word or phrase?

- A) undermine
- B) corroborate
- C) summarize
- D) restrict

Answer: B

Question 3 Skill: Words in Context

Stanley Kubrick's *2001: A Space Odyssey* is known for its glacially slow pacing. Iranian filmmaker Abbas Kiarostami even acknowledges that his own deliberately paced films sometimes cause audiences to lose focus. This might imply his films are tedious, but tedium does not always produce drowsiness: boredom generates a restless tension when the mind lacks a point of fixation, and thus it brings about not lethargy but _____.

Which choice completes the text with the most logical and precise word or phrase?

- A) resilience
- B) agitation
- C) apathy
- D) tranquility

Answer: B

Question 4 Skill: Words in Context

Challenges in recording the foraging behavior of nocturnal bats and a general research emphasis on daytime insect pollinators, such as *Apis mellifera*, have resulted in _____ of data on nighttime bat pollination of plants such as *Agave tequilana*, but Dr. Rodrigo Medellín et al. have addressed this by deploying acoustic monitoring devices.

Which choice completes the text with the most logical and precise word or phrase?

- A) a surplus
- B) a collection
- C) a scarcity
- D) a contradiction

Answer: C

Question 5 Skill: Function of Underlined Sentence

Vertical inheritance involves the passage of genetic traits from parent to offspring; horizontal transmission, by contrast, involves the transfer of traits between organisms not in a parent-offspring lineage. While horizontal transmission is well documented among prokaryotes — single-celled organisms such as the bacteria *Escherichia coli* and *Staphylococcus aureus* — it has seldom been confirmed in eukaryotes (typically multicellular organisms). **However, recent genomic analyses suggest that horizontal transmission is more prevalent in eukaryotes than previously recognized.**

Which choice best states the function of the underlined sentence in the text as a whole?

- A) It implies that a widely held assumption about horizontal transmission may be incorrect.
- B) It compares the rates at which horizontal transmission has been detected in two categories of organisms.
- C) It argues that a particular line of research concerning horizontal transmission is unlikely to yield results.
- D) It draws a distinction between horizontal transmission and vertical inheritance.

Answer: A

Question 6 Skill: Main Purpose

The Master and Margarita, first published in 1966, is a novel by Mikhail Bulgakov that blends satire, fantasy, and philosophical inquiry. Some literary scholars have praised the novel's abrupt, seemingly inconclusive chapter endings, arguing that they create deliberate disorientation befitting its fantastical narrative. But other scholars contend the chapters should not be read as deliberately fragmentary but as parts of a single, cohesive artistic vision, comparable in scope to Leo Tolstoy's *War and Peace*.

Which choice best states the main purpose of the text?

- A) To present a reason the unusual structure of *The Master and Margarita* makes it one of Bulgakov's most challenging works
- B) To argue that *The Master and Margarita* should have the same literary reputation as *War and Peace*
- C) To describe a characteristic of *The Master and Margarita* and offer two differing viewpoints on this characteristic
- D) To explain why many scholars find *The Master and Margarita* engaging despite perceived structural flaws

Answer: C

Question 7 Skill: Overall Structure

With its fusion of jazz harmonics and West African rhythmic traditions, *Bitches Brew* is representative of Miles Davis's recordings in the late 1960s and early 1970s. Through albums bridging these musical worlds, he achieved some of his most groundbreaking milestones. A decade earlier, however, Davis had first established himself as a virtuoso rooted in the bebop tradition of New York City, where he was born and raised. The influence of the city's jazz clubs can be heard in the improvisational approach of *Kind of Blue*, one of his most celebrated albums from this earlier period.

Which choice best describes the overall structure of the text?

- A) It examines Davis's lifelong commitment to jazz fusion, then debates whether bebop influenced his later work.
- B) It discusses a particular period of Davis's recording career, then considers the music of an earlier period.
- C) It enthusiastically recommends Davis's later recordings, then expresses reservations about his earlier work.
- D) It praises *Bitches Brew*, then argues that it is remarkably similar to an album Davis recorded earlier.

Answer: B

Question 8 Skill: Cross-Text Comparison

Text 1 Research examining what makes people feel a sense of community has largely concentrated on bonds formed within established groups — from neighbors to teammates. But sociologist Dr. Priya Kapoor and colleagues have shown that momentary exchanges with strangers can also foster a meaningful sense of belonging. Stronger feelings of inclusion were reported by participants who briefly chatted with a cashier than by those who completed the same transaction in silence.

Text 2 Studies of social belonging frequently employ a framework that maps an individual's connections across three tiers. The innermost tier represents one's closest relationships (e.g., a spouse), the next tier includes reliable but less intimate contacts (e.g., a coworker), and the outermost tier comprises acquaintances and peripheral contacts (those familiar enough to acknowledge but not to confide in).

Based on the texts, what would Kapoor and colleagues (Text 1) most likely say about the model in Text 2?

- A) It highlights that much research on belonging overlooks a type of interaction — exchanges with strangers — that can positively contribute to people's sense of inclusion.
- B) It reflects an overemphasis on relationship duration in researchers' assessments of the importance of various connections.
- C) It explains observations that individuals typically expect interactions with friends to be more positive than interactions with strangers.
- D) It emphasizes distinctions among types of close connections that are not adequately represented in belonging research.

Answer: A

Question 9 Skill: Shared Topic (Dual Passage)

Text 1 From the enormous blue whale to the tiny bumblebee bat, the diversity of mammals never ceases to fascinate biologists. Yet there is still much to discover. Unlike their terrestrial ancestors, modern cetaceans — whales and dolphins — spend their entire lives in the ocean. Their aquatic lifestyle makes it extremely difficult for scientists to observe them consistently.

Text 2 By attaching satellite-linked tags to individual humpback whales, marine biologist Dr. Ari Friedlaender and colleagues can now track previously hidden migration routes. Such tags can overturn long-standing beliefs. For decades it was assumed humpback whales traveled in a straight line between feeding and breeding grounds. But tagged whales often take winding, indirect paths spanning thousands of additional miles.

The author of Text 1 and the author of Text 2 both discuss which topic?

- A) How modern cetaceans compare with their terrestrial ancestors
- B) Scientists' efforts to study cetaceans
- C) The advantages of attaching satellite tags to marine mammals
- D) Changes in cetacean migration patterns over time

Answer: B

Question 10 Skill: Quantitative Inference

Baleen whales consume up to 30 percent of their total body mass in krill per day. So in one day, a fin whale weighing 50,000 kg could consume 15,000 kg, while a blue whale weighing 150,000 kg could ingest 45,000 kg. Over recent decades, certain baleen whale populations have recovered from near-extinction, and alongside this recovery, krill populations have paradoxically increased. Marine ecologist Dr. Victor Smetacek explains this by noting that baleen whales recycle iron through their fecal plumes, promoting phytoplankton blooms that sustain krill populations.

Based on the text, what can most reasonably be concluded about fin whales and blue whales?

- A) As krill consumption by fin whales has decreased, the krill consumption of blue whales has increased.
- B) The blue whale is capable of consuming more krill per day than the fin whale is.
- C) Both species eat krill, but only the fin whale can also feed on phytoplankton.
- D) Populations of both species have increased significantly due to rising iron levels in the ocean.

Answer: B

Question 11 Skill: Data Interpretation (Bar Chart)

[Bar chart: "Correlation between Model-Predicted and Listener-Reported Enjoyment Ratings, by Music Genre." Three bars each for Classical and Electronic. Classical: L4 $\approx$ 0.38, L1 $\approx$ 0.55, L2 $\approx$ 0.40. Electronic: L4 $\approx$ 0.25, L1 $\approx$ 0.23, L2 $\approx$ 0.22.]

Music psychologist Dr. Aiko Tanaka and colleagues built a computational model to predict how much a listener would enjoy a particular piece of music on a scale from 1 to 5, then had participants rate selections in various genres. Higher correlations indicate closer agreement between predictions and actual ratings. Assuming L1 and L2 gave equal ratings to the classical selections, the graph indicates that the model predicted _____.

Which choice most effectively uses data from the graph to complete the statement?

- A) L1's and L2's ratings for classical music would equal one another.
- B) L1's and L2's ratings for classical music would differ from one another.
- C) L1 would derive more aesthetic pleasure from classical music than from electronic music.
- D) L1 would derive less aesthetic pleasure from classical music than from electronic music.

Answer: C

Question 12 Skill: Textual Evidence (Quotation from Play)

A Raisin in the Sun is a 1959 play by Lorraine Hansberry. The Younger family lives in a cramped Chicago apartment and receives a \$10,000 insurance payment. Hansberry portrays the tension between Walter Lee and his mother Lena as deeply rooted in their conflicting visions for the family's future, as is evident when _____.

Which quotation from a translation or performance text of the play most effectively illustrates the claim?

- A) Lena tells Ruth, "When you starts measuring a man, measure him right, child."
- B) Walter Lee says to Lena, "I want so many things that they are driving me kind of crazy... sometimes I can see the future stretched out in front of me — just plain as day... all I got to do is reach out and grab it."
- C) Ruth says of Walter Lee, "He was out all night again last night... eats breakfast and goes out again."
- D) Lena says to Walter Lee, "I seen him night after night come in and look at that rug, and then look at me."

Answer: B

Question 13 Skill: Weaken Hypothesis

Some industrial processes produce titanium dioxide nanoparticles (TiO₂-NPs), which can enter waterways via runoff. In a 2020 study, Dr. Ingrid Holm demonstrated that TiO₂-NPs can accumulate in freshwater mussels (*Dreissena polymorpha*). It has been hypothesized that *D. polymorpha* could serve as a useful biological indicator for nanoparticle contamination, replacing the need for costly sampling of vertebrate species like rainbow trout (*Oncorhynchus mykiss*) in regulatory compliance testing.

Which finding, if true, would most directly weaken the hypothesis?

- A) Compared with *O. mykiss*, *D. polymorpha* can accumulate detectable TiO₂-NP concentrations with significantly fewer negative health effects.
- B) The rate of TiO₂-NP uptake in *D. polymorpha* differs from the rate in *O. mykiss* in a way that is not yet well understood.
- C) *D. polymorpha* has been shown to accumulate several other types of nanoparticles, whereas *O. mykiss* accumulates only TiO₂-NPs.
- D) When both species are exposed to similar TiO₂-NP levels, concentrations show significant individual-to-individual variation in both species.

Answer: A

Question 14 Skill: Strengthen Claim

Dr. Mariana Castellanos and colleagues investigated the function of sound production in silk moth and hawk moth caterpillars. They found that during simulated attacks, 33% of tested species produced sound, including chirps, whistles, vocalizations, and clicks. Species that also released a defensive chemical compound produced exclusively clicks and chirps — sounds that tend to be shorter in duration than whistles or vocalizations. Castellanos and colleagues argue that sounds produced by chemically defended species serve primarily as a warning to educate predators rather than to startle them.

Which finding, if true, would most directly support Castellanos and colleagues' claim?

- A) Short-duration sounds like clicks and chirps tend to be especially effective in training animals to associate a stimulus with a negative consequence.
- B) Caterpillars with chemical defenses produced sounds at higher volume than caterpillars without chemical defenses.
- C) The acoustic characteristics of clicks and chirps make them particularly audible to bats, lizards, and birds.
- D) Chickens and yellow warblers have been observed to stop their attacks in response to caterpillar sounds of any duration.

Answer: A

Question 15 Skill: Logical Completion (Art/Abstraction)

The compositional approach of *Composition VII*, a 1913 painting by Wassily Kandinsky, is closely aligned with early abstract art — a movement led primarily by European artists — rather than with traditional representational forms. Few viewers would deduce from *Composition VII* that Kandinsky was deeply influenced by Russian folk art. In this regard, the work typifies how European abstractionists incorporated non-Western influences in ways invisible to audiences. By contrast, contemporary Māori artist Shane Cotton assembles abstract forms derived from traditional kōwhaiwhai patterns and carved meeting-house motifs. Thus in Cotton’s work, abstraction has the effect of _____.

Which choice most logically completes the text?

- A) making Indigenous cultural references more accessible than they are in *Composition VII*.
- B) asserting the Indigenous origins of patterns associated with early European abstraction.
- C) challenging the dominance of European artists within the abstract art movement.
- D) bridging Indigenous and European traditions that coexist uneasily in *Composition VII*.

Answer: A

Question 16 Skill: Logical Completion (Archaeology)

Archaeologists generally hold that when a major population displacement disrupts the transmission of craft knowledge, this manifests as simultaneous declines in multiple specialized traditions. Inventories from excavation sites in the Indus River valley dating from 2600 to 1500 BCE document a marked reduction around 1900 BCE in objects featuring elaborate bead-drilling, an advanced lapidary technique. The inventories also show that sophisticated ceramic glazing and most other specialized traditions persisted through this period, a finding suggesting that _____.

Which choice most logically completes the text?

- A) disruptions in bead-drilling around 1900 BCE likely occurred alongside widespread population displacement.
- B) a sudden simultaneous decline across multiple crafting traditions is less likely than previously assumed to indicate that expertise transmission was disrupted by demographic changes.
- C) peoples in the Indus River valley continued to maintain multiple specialized crafting traditions, including glazed ceramics, even as demographic shifts occurred.
- D) cross-cultural exchange between distinct demographic groups likely explains the expansion of ceramic glazing around 1900 BCE.

Answer: C

Question 17 Skill: Verb Form

In a chapter titled “Beautiful Objects” from Lady Murasaki Shikibu’s *The Tale of Genji*, the author describes with delight a child’s silk robe, the glow of fireflies along a garden stream, and cherry blossoms scattering in the wind. Such catalogs, composed with elegant precision by Murasaki, who served as a lady-in-waiting to Empress Shōshi in the eleventh century, _____ the refined sensibility of Heian-era Japanese court culture.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) reveals
- B) was revealing
- C) to reveal
- D) reveal

Answer: D

Question 18 Skill: Punctuation (List with semicolons)

Referencing the example of Athena, a mythological goddess whose temple was allegedly discovered on the Acropolis of Athens, the historian Plutarch (first century CE) suggested that all the major figures in the

Greek religious tradition — including Poseidon, the god of the sea; Hermes, the messenger god; and Ares, the god of _____ may have been historical rulers who, through the power of legend, became deified.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) war;
- B) war—
- C) war,
- D) war

Answer: C

Question 19 Skill: Subject-Verb Agreement

Classified as a trochee, the word “raven” consists of one stressed syllable followed by one unstressed syllable. The word “divine,” by comparison, consists of one unstressed syllable followed by one stressed syllable, a metrical foot _____ to in English metrical verse as an iamb.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) referred
- B) is referred
- C) referring
- D) refers

Answer: B

Question 20 Skill: Comma with compound predicate

The green spaces of Portland, Oregon, appear to make residents happier. In a 2021 study, researchers investigating the relationship between outdoor environments and emotional well-being _____ geotagged social media posts from neighborhoods across Portland. They discovered that posts from near the city’s parks included more language associated with positive emotions than other posts did.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) analyzed and the post’s sentiment
- B) analyzed, and the post’s sentiment
- C) analyzed and the post’s sentiment,
- D) analyzed, and the post’s sentiment,

Answer: B

Question 21 Skill: Semicolons and commas

With a generating capacity of 22,500 megawatts, the Three Gorges hydroelectric dam in China produces an enormous amount of renewable energy. Power stations like Three Gorges supply only a small fraction of the world’s _____ the difficulty of finding locations with sufficient river flow and suitable geology limiting the widespread construction of such facilities.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) electricity; though,
- B) electricity, though;
- C) electricity, though
- D) electricity, though,

Answer: C

Question 22 Skill: Transitions

The album *Buena Vista Social Club* received widespread acclaim for its contribution to the son cubano genre, a style of Cuban music featuring intricate guitar work and layered percussion that originated in eastern Cuba in the late nineteenth century. _____ few were surprised when the album won the Grammy Award for Best Tropical Latin Album in 1998.

Which choice completes the text with the most logical transition?

- A) Frequently,
- B) Therefore,
- C) However,
- D) Granted,

Answer: B

Question 23 Skill: Transitions

In 1935, the U.S. Coast and Geodetic Survey calculated the geographic center of the contiguous United States using only a cardboard cutout and a plumb line. _____ this seemingly primitive approach produced a remarkably accurate result: when the 1935 calculation was compared to a center point computed by GPS over sixty years later, the two differed by only a handful of miles.

Which choice completes the text with the most logical transition?

- A) Still,
- B) Similarly,
- C) In other words,
- D) Accordingly,

Answer: A

Question 24 Skill: Transitions

In an ekphrastic poem, a poet responds to a work of visual art such as a sculpture, photograph, or painting. Natasha Trethewey's 2006 poem "Photograph: Ice Storm, 1971," for example, reflects on a family snapshot; Mark Doty's 1993 poem "Still Life with Oysters and Lemon" _____ draws its inspiration from a seventeenth-century Dutch painting.

Which choice completes the text with the most logical transition?

- A) nevertheless,
- B) conversely,
- C) likewise,
- D) hence,

Answer: B

Question 25 Skill: Rhetorical Synthesis (Provide Example)

While researching a topic, a student has taken the following notes:

- Sound engineers design and operate audio equipment for live events and recordings.
- They adjust microphone placement, mixing levels, and effects processing.
- *Thriller* (1982) was produced by Quincy Jones.
- Bruce Swedien was the album's lead sound engineer.

Which choice most effectively provides an example of an album a sound engineer worked on?

- A) Sound engineer Bruce Swedien collaborated with producer Quincy Jones.
- B) The 1982 album *Thriller* is one example of Bruce Swedien's work as a sound engineer.
- C) In music production, sound engineers work with microphones, mixing boards, and effects processors.
- D) As a sound engineer, Bruce Swedien adjusted microphone placement and mixing levels for recordings.

Answer: B

Question 26 Skill: Rhetorical Synthesis (Example of a concept)

While researching a topic, a student has taken the following notes:

- The Japanese word “ikigai” refers to a person’s sense of purpose in life.
- In many cultures, different words capture distinct philosophies about meaningful living.
- The Danish concept of “hygge” describes a feeling of coziness and comfortable conviviality.
- Denmark’s culture is known for its emphasis on work-life balance and communal warmth.

The student wants to provide an example of a culture whose language includes a word for a specific philosophy about well-being. Which choice most effectively accomplishes this goal?

- A) The Japanese word “ikigai” is the origin of a broader cultural interest in purpose-driven living.
- B) In many cultures, specific vocabulary captures distinct approaches to meaningful living.
- C) Denmark’s culture of work-life balance is reflected in the Danish word “hygge,” which describes a feeling of coziness and comfortable togetherness.
- D) Denmark does not use the Japanese term “ikigai” to describe its cultural emphasis on well-being.

Answer: C

Question 27 Skill: Rhetorical Synthesis (Emphasize Similarity)

While researching a topic, a student has taken the following notes:

- The ratio of the total mass of an element to the combined mass of all elements in an environment is called the mass fraction.
- The mass fraction of iron (Fe) on Earth is 319,000 parts per million (ppm).
- This indicates that the relative abundance of iron on Earth is 31.9%.
- The ratio of the total number of atoms of an element to the total number of all atoms in an environment is called the mole fraction.
- The mole fraction of iron on Earth is 18,200,000 parts per billion (ppb).
- This indicates that the relative abundance of iron on Earth is 1.82%.

The student wants to emphasize a similarity between mass fraction and mole fraction. Which choice most effectively accomplishes this goal?

- A) On Earth, the mass fraction of iron is 319,000 ppm, or 31.9%, and the mole fraction is 18,200,000 ppb, or 1.82%.
- B) While mass fraction measures relative mass and mole fraction measures relative atom count, both indicate the relative abundance of elements in an environment.
- C) Both mass and mole fraction express the abundance of iron on Earth as a ratio of iron to the combined total of all elements.
- D) The mass fraction of iron indicates its relative mass; in addition, the element’s mole fraction indicates the relative number of iron atoms.

Answer: B

SECTION 2, MODULE 1: Math

Question 1

A triangle has one side with a length of a centimeters and two sides each with a length of b centimeters. The perimeter of the triangle is 53 centimeters. Which equation represents this situation?

- A) $a + b = 53$
- B) $a + 2b = 53$
- C) $2a + b = 53$

D) $2a + 2b = 53$

Answer: B

Question 2

If $5 + x = 3$, what is the value of $30 + 6x$?

Answer: 18

Question 3

A recipe for trail mix uses 3 cups of granola and 5 handfuls of almonds and contains a total of 1,710 milligrams (mg) of potassium. There are 420 mg of potassium in 1 cup of granola. How much potassium, in mg, is in 1 handful of almonds?

- A) 1,290
- B) 516
- C) 90
- D) 18

Answer: C

Solution: Total from granola = $3 \times 420 = 1,260$ mg. Remaining = $1,710 - 1,260 = 450$ mg. Per handful = $450 \div 5 = 90$ mg.

Question 4

An electronics store is offering all headphones at a 15% discount off their original price for the month of November. What is the discount, in dollars, on headphones with an original price of \$80?

- A) 68
- B) 65
- C) 20
- D) 12

Answer: D

Solution: $0.15 \times 80 = 12$.

Question 5

What is the value of $3(2x - 4)$ when $x = 5$?

- A) 6
- B) 14
- C) 18
- D) 26

Answer: C

Solution: $3(10 - 4) = 3(6) = 18$.

Question 6

$$\begin{aligned} 3x &= 21 \\ -2x + y &= -9 \end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

- A) -14
- B) -5
- C) 5
- D) 12

Answer: D

Solution: $x = 7$. Then $-14 + y = -9$, so $y = 5$. $x + y = 12$.

Question 7

A customer is buying apples that cost \$0.85 each. If the customer has a coupon for \$3 off the entire purchase and there is no sales tax, which inequality represents the number of apples, x , that the customer can purchase for no more than \$15?

- A) $0.85x + 3 \leq 15$
- B) $0.85x + 3 \geq 15$
- C) $0.85x - 3 \leq 15$
- D) $0.85x - 3 \geq 15$

Answer: C

Solution: Cost after coupon = $0.85x - 3$. Must be ≤ 15 .

Question 8

Circle R has a radius of 25 millimeters (mm). What is the area of circle R, in mm^2 ?

- A) 25π
- B) 50π
- C) 100π
- D) 625π

Answer: D

Solution: $A = \pi r^2 = \pi(25)^2 = 625\pi$.

Question 9

In the xy -plane, circle P is defined by the equation $(x + 5)^2 + (y + 3)^2 = 25$. Circle Q has the same center as circle P, and the radius of circle Q is 3 units greater than the radius of circle P. Circle Q is defined by the equation $(x + 5)^2 + (y + 3)^2 = p$, where p is a constant. What is the value of p ?

Answer: 64

Solution: Radius of P = 5. Radius of Q = $5 + 3 = 8$. $p = 8^2 = 64$.

Question 10

Which expression is equivalent to $5ab^3 + 2ab + 3ab^3$?

- A) $8ab^3 + 2ab$
- B) $8ab^6 + 2ab$
- C) $10ab^6$
- D) $10ab^7$

Answer: A

Solution: Combine like terms: $(5ab^3 + 3ab^3) + 2ab = 8ab^3 + 2ab$.

Question 11

If $x/y = 84$ and $cx/(4y) = 84$, what is the value of c ?

Answer: 4

Solution: $cx/(4y) = c/4 \cdot (x/y) = c/4 \cdot 84 = 84$. So $c/4 = 1$, $c = 4$.

Question 12

In the figure shown, triangle PQR is similar to triangle PST. The measure of angle PST is 63° , and $QR = 15(ST)$. What is the measure of angle PQR?

- A) $(15 \cdot 63)^\circ$
- B) $(15 + 63)^\circ$
- C) 63°
- D) 15°

Answer: C

Solution: Corresponding angles in similar triangles are equal.

Question 13

The graph shows the estimated melting point y , in degrees Celsius, of a straight-chain fatty acid with a carbon chain length of x atoms, where $4 \leq x \leq 24$. Which statement is the best interpretation of the point $(16.5, 62.8)$?

- A) A fatty acid with a chain length of 62.8 carbons has an estimated melting point of 16.5°C .
- B) A fatty acid with a chain length of 16.5 carbons has an estimated melting point of 62.8°C .
- C) The minimum estimated melting point corresponds to 16.5 carbons and 62.8°C .
- D) The maximum estimated melting point corresponds to 16.5 carbons and 62.8°C .

Answer: B

Question 14

The function g is defined by $g(x) = 2x^2 - 8x + 5$. What is the value of $g(3)$?

- A) -1
- B) 0
- C) 1
- D) 5

Answer: A

Solution: $g(3) = 2(9) - 8(3) + 5 = 18 - 24 + 5 = -1$.

Question 15

If $84/x = 7x$, what is the value of $7x^2$?

- A) 504
- B) 84
- C) 12
- D) 7

Answer: B

Solution: $84/x = 7x \rightarrow 84 = 7x^2 \rightarrow 7x^2 = 84$.

Question 16

$$y \leq 3x - 14$$

Which point (x, y) is a solution to the given inequality in the xy -plane?

- A) $(-3, -14)$
- B) $(0, 14)$
- C) $(3, -14)$
- D) $(4, 14)$

Answer: C

Solution: Check $(3, -14)$: $-14 \leq 3(3) - 14 = -5$. Since $-14 \leq -5$ is true, $(3, -14)$ works.

Question 17

A gardener is planting zinnia seeds and cosmos seeds in a flower bed. The gardener will plant x zinnia seeds and y cosmos seeds. The gardener wants to plant at least 10 seeds total and spend no more than \$24. Each zinnia seed costs \$2, and each cosmos seed costs \$3. Which system of inequalities represents this situation?

- A) $x + y \geq 10$ and $2x + 3y \leq 24$
- B) $x + y \leq 10$ and $2x + 3y \leq 24$
- C) $x + y \geq 10$ and $3x + 2y \leq 24$
- D) $x + y \leq 10$ and $3x + 2y \geq 24$

Answer: A

Question 18

The given equation relates the positive numbers f , x , and g :

$$f = \frac{(8x)^2}{5g}$$

Which equation correctly expresses x in terms of f and g ?

- A) $x = \sqrt{5fg} / 8$
- B) $x = 8\sqrt{5fg}$
- C) $x = \sqrt{5fg / 8}$
- D) $x = \sqrt{5fg} - 8$

Answer: A

Solution: $f = 64x^2/(5g) \rightarrow 5fg = 64x^2 \rightarrow x^2 = 5fg/64 \rightarrow x = \sqrt{5fg}/8$.

Question 19

In the xy -plane, line m passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 4$. If line m also passes through the point $(6, d)$, what is the value of d ?

Answer: 48

Solution: Parallel lines have the same slope. Line m : $y = 8x$ (through origin). At $x = 6$: $y = 48$.

Question 20

A circle in the xy -plane has its center at $(14, 19)$ and has a radius of $7k$. Which equation represents this circle?

- A) $(x - 14)^2 + (y - 19)^2 = 49k$
- B) $(x - 14)^2 + (y - 19)^2 = 49k^2$
- C) $(x - 14)^2 + (y - 19)^2 = 7k$
- D) $(x - 14)^2 + (y - 19)^2 = 7k^2$

Answer: B

Solution: Standard form: $(x - h)^2 + (y - k)^2 = r^2$. $r = 7k$, so $r^2 = 49k^2$.

Question 21

According to a U.S. building code, ramps for public wheelchair access must form an angle θ with level ground such that $\tan \theta \leq 1/12$. If a ramp conforms to this code and has a height of 29.1 inches, what is the least possible value of the horizontal distance x , in inches?

Answer: 349.2

Solution: $\tan \theta = \text{height}/x$. For the maximum allowed angle: $1/12 = 29.1/x \rightarrow x = 29.1 \times 12 = 349.2$.

Question 22

The equation of a graph is $y = a^x + b$, where a and b are constants. The graph passes through the points $(0, 0)$ and $(1, 2)$. What is the value of $a - b$?

Answer: 4

Solution: At $(0, 0)$: $0 = 1 + b \rightarrow b = -1$. At $(1, 2)$: $2 = a - 1 \rightarrow a = 3$. $a - b = 3 - (-1) = 4$.

Question 1

$$f(x) = \frac{x + 9}{3}$$

For the function f defined by the equation shown, what is the value of $f(-6)$?

- A) -3
- B) -1
- C) 1
- D) 3

Answer: C

Solution: $f(-6) = (-6 + 9)/3 = 3/3 = 1$.

Question 2

28, 28, 29, 30, 32, 33, 37

What is the mean of the data set shown?

- A) 28
- B) 30
- C) 31
- D) 37

Answer: C

Solution: $(28 + 28 + 29 + 30 + 32 + 33 + 37) = 217$. $217 \div 7 = 31$.

Question 3

The voltage across a capacitor in an electric circuit is initially 16 volts and decreases with time, as shown in a graph. Which of the following is closest to the time, in seconds, from the initial measurement at which the voltage is 15% of its initial value?

- A) 0.43
- B) 0.85
- C) 1.71
- D) 3.41

Answer: C

Reasoning: 15% of 16 = 2.4 volts. From the exponential decay curve, this occurs at approximately 1.71 seconds.

Question 4

A scatterplot shows the relationship between two variables, x and y , for data set A. A line of best fit is also shown, with a y -intercept of approximately 54.57. Data set B is created by subtracting 9 units from the value of y for each data point from data set A. Which of the following is closest to the y -coordinate of the y -intercept of the line of best fit for data set B?

- A) 45.57
- B) 53
- C) 59.62
- D) 62.75

Answer: A

Solution: Subtracting 9 from every y -value shifts the entire line down by 9. New y -intercept $\approx 54.57 - 9 = 45.57$.

Question 5

In the linear function f , $f(0) = 5$ and $f(7) = 5$. Which equation defines f ?

- A) $f(x) = 0$
- B) $f(x) = 7$
- C) $f(x) = 5$
- D) $f(x) = x + 5$

Answer: C

Solution: Since $f(0) = f(7) = 5$, the function is constant: $f(x) = 5$.

Question 6

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$.

x	$f(x)$
1	-20
5	-40
10	-65

Which equation defines f ?

- A) $f(x) = -5(x - 3)$
- B) $f(x) = -5(5x - 3)$
- C) $f(x) = -5(x + 3)$

D) $f(x) = -5(x - 4)$

Answer: C

Solution: Slope = $(-40 - (-20))/(5 - 1) = -20/4 = -5$. Using point $(1, -20)$: $-20 = -5(1) + b \rightarrow b = -15$. $f(x) = -5x - 15 = -5(x + 3)$. Check: $-5(1 + 3) = -20$ ✓. $-5(5 + 3) = -40$ ✓. $-5(10 + 3) = -65$ ✓.

Question 7

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 128(0.5)^{(t/29)}$ gives the approximate mass, in grams, remaining t years after a 128-gram sample starts to decay. Which statement is the best interpretation of $s(58) = 32$ in this context?

- A) Approximately 32 grams of the sample remains 58 years after the sample starts to decay.
- B) The mass has decreased by approximately 32 grams 58 years after the sample starts to decay.
- C) The mass has decreased by approximately 58 grams 32 years after the sample starts to decay.
- D) Approximately 58 grams of the sample remains 32 years after the sample starts to decay.

Answer: A

Solution: $s(58) = 32$ means that when $t = 58$ years, 32 grams remain.

Question 8

The table shows the distribution of people in a certain city by age group.

Age group	Proportion
Less than 18 years old	34%
18-40 years old	23%
41-65 years old	22%
Greater than 65 years old	21%

If a person in this city is selected at random, which of the following is closest to the probability of selecting a person who is greater than 65 years old, given that the person is at least 18 years old?

- A) 0.21
- B) 0.32
- C) 0.45
- D) 0.62

Answer: B

Solution: $P(\text{at least } 18) = 23 + 22 + 21 = 66\%$. $P(>65 \text{ and } \geq 18) = 21\%$. $P(>65 | \geq 18) = 21/66 \approx 0.318 \approx 0.32$.

Question 9

A right circular cone has a base diameter of 28 inches and a height of 6 inches. The volume of this cone is $k\pi$ cubic inches. What is the value of k ?

Answer: 392

Solution: $r = 14$. $V = (1/3)\pi r^2 h = (1/3)\pi(196)(6) = (1/3)\pi(1176) = 392\pi$. So $k = 392$.

Question 10

In triangle ABC, point D is the midpoint of segment AB and point E is the midpoint of segment BC. If $BD = 8$ and $DE = 14$, what is the length of segment AC?

- A) 28
- B) 26
- C) 22
- D) 16

Answer: A

Solution: By the triangle midsegment theorem, DE is parallel to AC and $DE = AC/2$. So $AC = 2 \times 14 = 28$.

Question 11

$$x(r - 7) + 3 = 19x + 25$$

In the given equation, r is a positive integer. If the equation has exactly one solution, what CANNOT be the value of r ?

- A) 3
- B) 7
- C) 22
- D) 26

Answer: D

Solution: Expand: $xr - 7x + 3 = 19x + 25 \rightarrow xr - 7x - 19x = 25 - 3 \rightarrow x(r - 26) = 22$. If $r = 26$, then $0 = 22$, which is impossible (no solution). For exactly one solution, $r \neq 26$. The value that CANNOT work is 26.

Question 12

In a collection of items, 10% are red, 25% are green, 35% are blue, and 30% are yellow. If there are 40 green items, how many blue items are in the collection?

- A) 14
- B) 16
- C) 48
- D) 56

Answer: D

Solution: $25\% = 40 \rightarrow \text{total} = 160$. Blue = 35% of 160 = 56.

Question 13

$$x(13x + 27) + c = 5$$

In the given equation, c is a constant. A solution to the equation is $(-27 + \sqrt{261}) / 26$. What is the value of c ?

- A) 14
- B) 9
- C) 5
- D) 4

Answer: A

Solution: Expand: $13x^2 + 27x + c - 5 = 0$. By the quadratic formula: $x = (-27 \pm \sqrt{(729 - 52(c - 5))}) / 26$. The given solution has $\sqrt{261}$, so: $729 - 52(c - 5) = 261 \rightarrow 52(c - 5) = 468 \rightarrow c - 5 = 9 \rightarrow c = 14$.

Question 14

One of the factors of $3x^3 + 48x^2 + 180x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b ?

Answer: 6

Solution: Factor out $3x$: $3x(x^2 + 16x + 60) = 3x(x + 6)(x + 10)$. The factors with form $(x + b)$ where $b > 0$ are $(x + 6)$ and $(x + 10)$. Smallest $b = 6$.

Question 15

A model estimates that at the end of each year from 2015 to 2020, the number of rabbits in a population was 170% more than the number at the end of the previous year. The model estimates that at the end of 2016, there were 216 rabbits. Which equation represents this model, where n is the estimated number of rabbits t years after the end of 2015 and $t \leq 5$?

- A) $n = 80(1.7)^t$
- B) $n = 80(2.7)^t$
- C) $n = 216(1.7)^t$
- D) $n = 216(2.7)^t$

Answer: B

Solution: 170% more means $\times 2.7$ each year. At end of 2015 ($t = 0$): $n_0 = 216/2.7 = 80$. So $n = 80(2.7)^t$.

Question 16

The figure shows a rectangular pool surrounded by a concrete path that is x feet wide on all sides. The pool is 21 ft long and 13 ft wide. The area of the concrete path is 240 ft^2 . What is the value of x ?

- A) 3
- B) 6
- C) 20
- D) 40

Answer: A

Solution: Total area = $(21 + 2x)(13 + 2x)$. Pool area = 273. Path area = $(21 + 2x)(13 + 2x) - 273 = 240$. So $(21 + 2x)(13 + 2x) = 513$. $273 + 42x + 26x + 4x^2 = 513 \rightarrow 4x^2 + 68x + 273 = 513 \rightarrow 4x^2 + 68x - 240 = 0 \rightarrow x^2 + 17x - 60 = 0 \rightarrow (x + 20)(x - 3) = 0$. $x = 3$ (positive value).

Question 17

The table shows three values of x and their corresponding values of y , where s is a constant. There is a linear relationship between x and y .

x	y
$-2s$	17
$-s$	14
s	8

Which equation represents this relationship?

- A) $sx + 3y = 11s$
- B) $3x + sy = 11s$
- C) $3x + sy = 11$
- D) $sx + 3y = 11$

Answer: B

Solution: Slope = $(14 - 17)/(-s - (-2s)) = -3/s$. Also $(8 - 14)/(s - (-s)) = -6/(2s) = -3/s$. ✓ Using point $(s, 8)$: $y - 8 = (-3/s)(x - s) \rightarrow y = (-3/s)x + 3 + 8 \rightarrow y = (-3/s)x + 11$. Multiply by s : $sy = -3x + 11s \rightarrow 3x + sy = 11s$.

Question 18

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?

Answer: -7

Solution: Four consecutive odd integers: $x, x+2, x+4, x+6$. Condition: $28(x + 4) \leq (x + x + 6) - 50$. $28x + 112 \leq 2x - 44$. $26x \leq -156$. $x \leq -6$. Since x must be odd: $x \leq -7$. Greatest odd value = -7 .

Question 19

A square map has a side length of 30 inches, and 1 inch on the map represents an actual distance of 17 miles. A smaller version of the same map is printed as a square with the side length 65% shorter than that of the original. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

- A) 5.95
- B) 10.30
- C) 19.50
- D) 48.57

Answer: D

Solution: Original side = 30 in. Total distance represented = $30 \times 17 = 510$ miles. Smaller side = $30 \times (1 - 0.65) = 30 \times 0.35 = 10.5$ in. Scale on smaller map: $510/10.5 \approx 48.57$ miles per inch.

Question 20

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$, and $b < 0$.

I. $f(x) = a(2.1)^{(x+b)}$ II. $g(x) = a(2.1)^x + b$

If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

Answer: B

Solution: For f : $f(0) = a(2.1)^b$ — this is not directly visible as a constant in the equation. For g : $g(0) = a(2.1)^0 + b = a + b$ — the y -intercept contains both a and b , and b is explicitly a constant/coefficient in the equation form $g(x) = a(2.1)^x + b$ (where $+b$ is the vertical shift, visible as a constant).

Question 21

The function h is defined by $h(x) = -\sqrt{x^2 + bx + c}$, where b and c are constants. In the xy -plane, the graph of $y = h(x)$ contains the points $(2, 0)$ and $(0, -\sqrt{358})$. If $h(m) = 0$, what is the greatest possible value of m ?

Answer: 2

Solution: $h(m) = 0$ requires the expression under the square root to equal zero. Using the given conditions $h(2) = 0$ and $h(0) = -\sqrt{358}$, solve for b and c , then find all zeros of $x^2 + bx + c = 0$. The greatest value of m satisfying $h(m) = 0$ is 2.

Question 22

The function f is defined by $f(x) = |x|/a - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?

Answer: 638

Solution: Since $a < 0$, $|a| = -a$. $f(15a) = |15a|/a - 14 = 15|a|/a = 15(-a)/a = -15$. So $f(15a) = -15 - 14 = -29$. $f(8a) = |8a|/a - 14 = 8(-a)/a = -8$. So $f(8a) = -8 - 14 = -22$. Product = $(-29)(-22) = 638$.

End of Full Practice Test

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